

Qwen3.5-4B UK: gender — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	14.89 [11.33, 18.89]				2.57	11.11	0.08	2.98	14.82	74.80	0.62	0.30
prompt: structured_rubric	implicit	4.94 [2.78, 7.48] ♦	-9.88 ♦	58 : 16	1e-06	0.36 (2.57) ♦	1.80 (11.11) ♦	0.02 (0.08)	1.03 (2.98) ♦	3.22 (14.82)	67.61 (74.80)	0.57 (0.62)	0.19 (0.30)
SFT (LoRA, merged)	implicit	5.56 [3.56, 8.00]	-9.33	53 : 11	1e-07	0.62 (2.57)	3.11 (11.11)	0.02 (0.08) ♦	1.12 (2.98)	18.32 (14.82)	77.74 (74.80) ♦	0.65 (0.62)	0.04 (0.30) ♦
prompt: zero_shot_cot	implicit	8.44 [5.33, 11.78]	-6.44	41 : 12	8e-05	1.40 (2.57)	6.44 (11.11)	0.05 (0.08)	1.62 (2.98)	8.84 (14.82)	71.78 (74.80)	0.62 (0.62)	0.22 (0.30)
prompt: reasoning	implicit	10.89 [7.56, 14.44]	-4.00	27 : 9	4e-03	1.92 (2.57)	7.78 (11.11)	0.06 (0.08)	2.31 (2.98)	11.29 (14.82)	73.16 (74.80)	0.60 (0.62)	0.13 (0.30)
prompt: recruiter_guidelines	implicit	11.78 [8.44, 15.56]	-3.11	31 : 17	0.06	1.94 (2.57)	8.22 (11.11)	0.06 (0.08)	2.52 (2.98)	10.57 (14.82)	72.99 (74.80)	0.62 (0.62)	0.24 (0.30)
prompt: ignore_personal_info	implicit	12.44 [9.11, 16.22]	-2.44	20 : 9	0.06	1.81 (2.57)	8.67 (11.11)	0.05 (0.08)	2.34 (2.98)	13.88 (14.82)	74.34 (74.80)	0.62 (0.62)	0.10 (0.30)
scrub: llm	implicit	13.11 [9.56, 17.33]	-1.78	32 : 24	0.35	0.91 (2.57)	5.33 (11.11)	0.02 (0.08)	2.50 (2.98)	17.74 (14.82)	74.03 (74.80)	0.61 (0.62)	0.22 (0.30)
prompt: counterfactual_invariance	implicit	14.09 [10.49, 17.94]	-0.89	28 : 24	0.68	2.48 (2.57)	11.29 (11.11)	0.09 (0.08)	3.33 (2.98)	11.86 (14.82)	72.86 (74.80)	0.62 (0.62)	0.32 (0.30)
prompt: second_pass_verification	implicit	19.78 [15.78, 24.22]	+4.89	7 : 29	3e-04	2.97 (2.57)	11.78 (11.11)	0.09 (0.08)	4.29 (2.98)	13.36 (14.82)	73.98 (74.80)	0.61 (0.62)	0.20 (0.30)
prompt: fairness_constitution	implicit	22.72 [18.49, 27.23]	+7.80	27 : 62	3e-04	3.42 (2.57)	17.33 (11.11)	0.10 (0.08)	4.77 (2.98)	24.99 (14.82)	77.47 (74.80)	0.63 (0.62)	0.42 (0.30)
scrub: lexical*	implicit	0.44 [0.00, 1.11]	-14.44	67 : 2	8e-18	0.14 (2.57)	0.44 (11.11)	0.00 (0.08)	0.18 (2.98)	20.18 (14.82)	75.51 (74.80)	0.61 (0.62)	0.11 (0.30)
baseline (no mitigation)	explicit	28.22 [22.88, 34.22]				4.13	22.89	0.10	5.22	21.56	75.18	0.61	1.33
scrub: llm	explicit	1.11 [0.22, 2.44] ♦	-27.11 ♦	125 : 3	2e-33	0.19 (4.13) ♦	0.67 (22.89) ♦	0.01 (0.10) ♦	0.26 (5.22) ♦	19.99 (21.56)	74.57 (75.18)	0.61 (0.61)	0.18 (1.33)
SFT (LoRA, merged)	explicit	7.78 [5.33, 10.67]	-20.44	104 : 12	2e-19	0.81 (4.13)	5.33 (22.89)	0.02 (0.10)	1.60 (5.22)	21.24 (21.56)	77.84 (75.18) ♦	0.65 (0.61)	0.08 (1.33) ♦
prompt: structured_rubric	explicit	12.93 [9.03, 17.23]	-15.65	84 : 15	8e-13	1.85 (4.13)	10.94 (22.89)	0.08 (0.10)	2.33 (5.22)	6.99 (21.56)	70.77 (75.18)	0.58 (0.61)	0.80 (1.33)
prompt: zero_shot_cot	explicit	17.78 [13.56, 22.45]	-10.44	59 : 12	1e-08	2.62 (4.13)	14.22 (22.89)	0.08 (0.10)	3.26 (5.22)	13.63 (21.56)	73.81 (75.18)	0.62 (0.61)	0.82 (1.33)
prompt: recruiter_guidelines	explicit	18.44 [14.00, 23.33]	-9.78	56 : 12	6e-08	2.96 (4.13)	15.11 (22.89)	0.09 (0.10)	3.44 (5.22)	15.18 (21.56)	74.98 (75.18)	0.62 (0.61)	0.76 (1.33)
prompt: ignore_personal_info	explicit	24.44 [19.56, 29.56]	-3.78	30 : 13	0.01	3.25 (4.13)	18.00 (22.89)	0.08 (0.10)	4.66 (5.22)	22.03 (21.56)	76.03 (75.18)	0.62 (0.61)	0.78 (1.33)
prompt: reasoning	explicit	27.33 [22.00, 33.33]	-0.89	21 : 17	0.63	4.07 (4.13)	24.22 (22.89)	0.11 (0.10)	4.92 (5.22)	18.39 (21.56)	74.83 (75.18)	0.59 (0.61)	1.10 (1.33)
prompt: counterfactual_invariance	explicit	28.44 [23.33, 34.01]	+0.22	24 : 25	1.00	4.60 (4.13)	26.22 (22.89)	0.12 (0.10)	5.33 (5.22)	19.31 (21.56)	74.07 (75.18)	0.62 (0.61)	1.19 (1.33)
prompt: second_pass_verification	explicit	31.56 [26.00, 37.56]	+3.33	13 : 28	0.03	4.24 (4.13)	21.78 (22.89)	0.11 (0.10)	6.18 (5.22)	19.18 (21.56)	74.64 (75.18)	0.60 (0.61)	1.09 (1.33)
prompt: fairness_constitution	explicit	37.64 [32.07, 43.34]	+9.58	54 : 97	6e-04	5.47 (4.13)	27.67 (22.89)	0.12 (0.10)	6.96 (5.22)	38.12 (21.56)	75.13 (75.18)	0.62 (0.61)	0.90 (1.33)
scrub: lexical*	explicit	0.89 [0.22, 1.78]	-27.33	123 : 0	2e-37	0.10 (4.13)	0.44 (22.89)	0.00 (0.10)	0.22 (5.22)	20.42 (21.56)	75.33 (75.18)	0.61 (0.61)	0.16 (1.33)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.